Mitigating Example Order Effects in Lightweight Language Models

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Abstract

- Standard few-shot learning in sub-2B models is highly volatile, with performance fluctuating based solely on the arrangement of demonstrations.
- Lightweight models like Llama 3.2 1B exhibit a 36.42% Pairwise Disagreement Rate, frequently anchoring predictions to whichever label appears first.
- This "Reliability Gap" prevents small-scale LLMs from being used for consistent, automated linguistic analysis in edge-deployment scenarios.

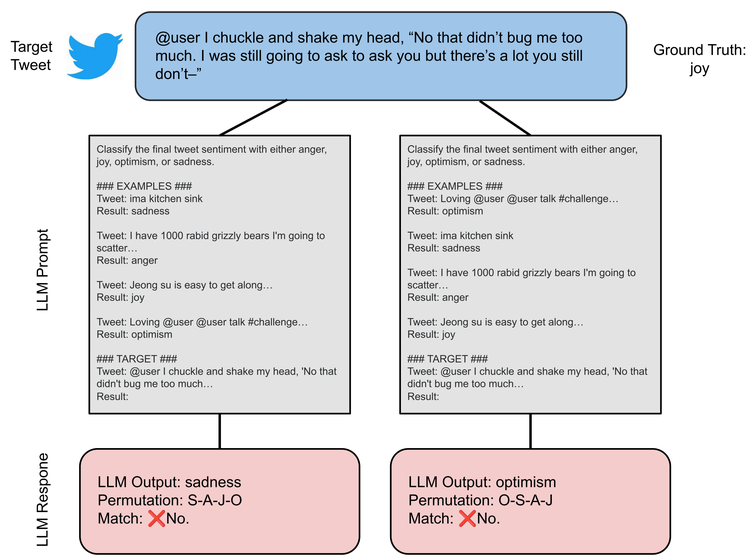
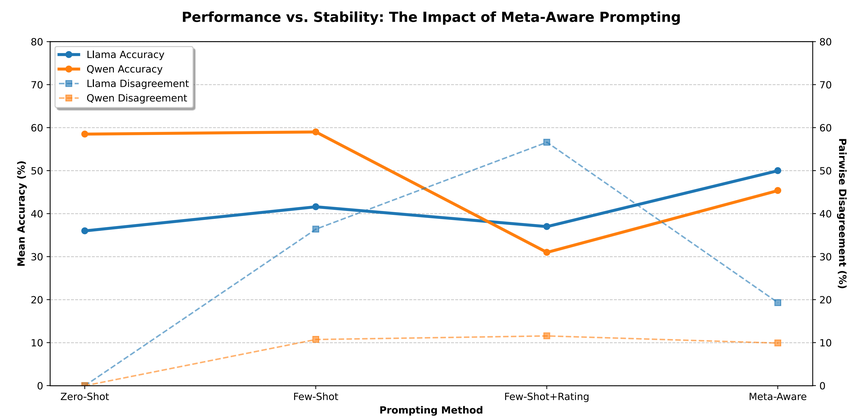


Figure 1: Demonstration of Order Sensitivity in Few-Shot Emotion Classification. The identical target tweet results in contradictory predictions when demonstration sequences are permuted, highlighting the reliability gap in lightweight LLMs. Permutation codes refer to the order of demonstrations (A: Anger, J: Joy, O: Optimism, S: Sadness).

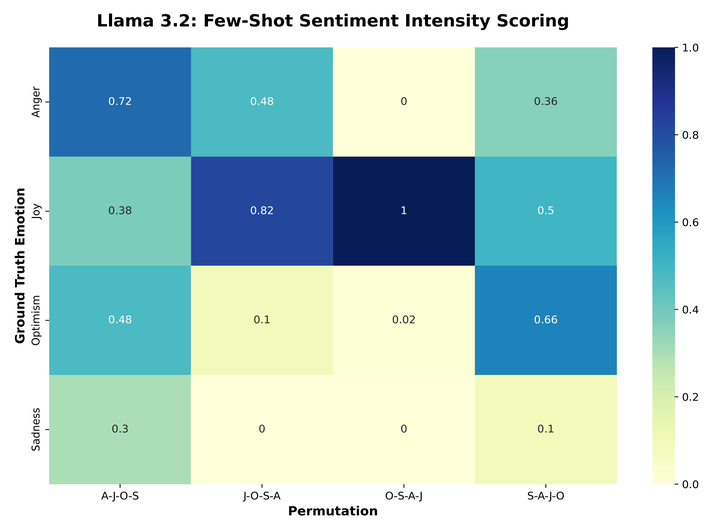
Methodology

- Utilized the TweetEval emotion subtask (*Anger, Joy, Optimism, Sadness*), sampling a balanced evaluation subset of 200 tweets.
- Each sample was tested against four cyclic permutations of few-shot demonstrations to measure Pairwise Disagreement Rate (PDR).
- Implemented Bias-Aware Meta-Prompting, which uses explicit positional warnings to decouple model predictions from example order.

Results



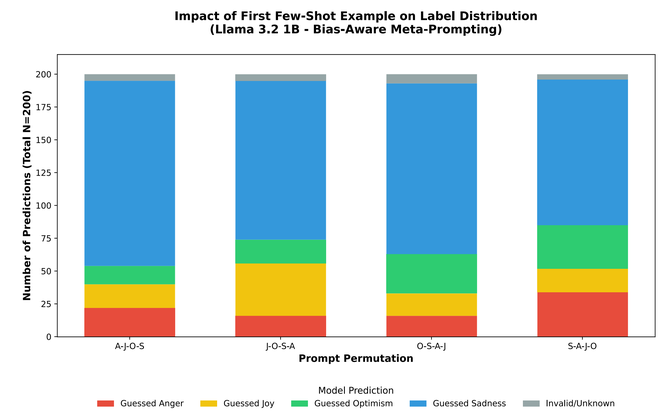
- Standard few-shot and rating-based methods trigger a spike in disagreement, indicating "haphazard" accuracy gains.
- Our method reverses this trend, stabilizing the model while reaching peak performance.
- Llama shows high "plasticity" (big moves), whereas Qwen demonstrates "rigidity" (stable but harder to influence).



- Incorporating a "1-10" confidence rating per emotion consistently degraded performance across both models.
- Small-scale models struggle to balance the dual-task requirement of sentiment analysis and numerical scoring, leading to increased hallucination and invalid responses
- Instead of stabilizing the model, the ranking instructions acted as a distractor, causing Llama and Qwen to ignore semantic evidence in favor of repetitive numerical patterns

Conclusion

- Bias-aware instructions allow Llama to prioritize semantic evidence over positional patterns, resulting in a 17.17 point reduction in disagreement



- Models still exhibit residual "label favorites" regardless of order. As seen in the above figure, Llama 3.2 shows a persistent skew toward "Sadness" despite structural intervention. Meanwhile, Qwen had a bias towards the "Anger" emotion.
- Qwen achieved its lowest PDR (9.9%) under the Bias-Aware strategy, yet it suffered a higher difference in accuracy between the best and worst performing model, a difference of 7.5%.
- The reduction in PDR across both models demonstrates a successful reduction the effects of example order on predictions. However, the plateau in Mean Accuracy suggests that structural consistency does not inherently guarantee semantic correctness.

Future Work

- Future research will explore the transferability of these structural prompts across broader reasoning tasks, such as logical deduction and mathematical problem solving.
- We aim to investigate whether selecting demonstrations based on semantic similarity, rather than fixed permutations, can further enhance the stabilizing effects of meta-aware scaffolding